

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

- First Line Contains Integer n – Size of array
- Next n lines Contains n numbers – Elements of an array
- Last Line Contains Integer x – Sum Value

Output Format

- First Line Contains Integer – Element1
- Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findPair(int arr[], int left, int right, int x, int *a, int *b) {
4      if (left >= right)
5          return 0;
6
7      int sum = arr[left] + arr[right];
8
9      if (sum == x) {
10         *a = arr[left];
11         *b = arr[right];
12         return 1;
13     }
14
15     if (sum < x)
16         return findPair(arr, left + 1, right, x, a, b);
17     else
18         return findPair(arr, left, right - 1, x, a, b);
19 }
20
21 int main() {
22     int n, x;
23     scanf("%d", &n);
24
25     int arr[n];
26
27     for (int i = 0; i < n; i++) {
28         scanf("%d", &arr[i]);
29     }
30
31     scanf("%d", &x);
32
33     int a, b;
34
35     if (findPair(arr, 0, n - 1, x, &a, &b)) {
36         printf("%d\n", a);
37         printf("%d\n", b);
38     } else {
39         printf("No\n");
40     }
41
42     return 0;
43 }
```

	Input	Expected	Got	
✔	4	4	4	✔
	2	10	10	
	4			
	8			
	10			
	14			
✔	5	No	No	✔
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✔